

**Institute of Mathematical Sciences
University of Malaya**

**SIM1001 Basic Mathematics
SIX1013 Fundamental of Advanced Mathematics
TUTORIAL 4 (Part I)**

1. Find real numbers x and y such that
 - (i) $(x + 2y) + (x - y)i = 1 + 4i$;
 - (ii) $(x + yi) + (2 + i) = 4 - i$.
2. Let $w = 1 + 2i$.
 - (a) Evaluate w^3 and w^{-3} .
 - (b) If w is a solution of the equation $z^3 + az + b = 0$ ($a, b \in \mathbb{R}$), find a and b , and show that $\bar{w}$ is also a solution.
3. Let $z_1 = 2 + i$, $z_2 = 3 - 2i$ and $z_3 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$.
Evaluate each of the following.
 - (i) $\left| \frac{z_1 + 2z_2 - 5 - i}{2z_1 - z_2 + 3 - i} \right|$
 - (ii) z_3^3 and $|z_3^4|$.
4. Express each of the following complex numbers
 - (i) in the polar form,
 - (ii) exponential form :
 - (a) $3 + 3i$
 - (b) $-1 + \sqrt{3}i$
 - (c) $4i$
 - (d) $\frac{2}{1 + i}$
5. Simplify each of the following complex numbers and write the answer in Cartesian form:
 - (a) $\frac{6 e^{i\pi/2}}{2 e^{i\pi/4}}$
 - (b) $8 e^{i\pi} (-2 e^{i2\pi})$
 - (c) $\left(\sqrt{3} e^{i\frac{2\pi}{3}} \right)^3$.

6. Find the modulus and argument of each of the following complex numbers:

(a) $w = \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)^3 \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)^2$

(b) $z = \frac{-2 \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)}{\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)}$

7. Evaluate (a) $(-1 + \sqrt{3}i)^{100}$ (b) $\left(\frac{2}{1+i} \right)^{20}$

8. Find (a) the 4th-roots of unity;

(b) the cube roots of $-1 + \sqrt{3}i$.

9. Solve each of the following equations and locate the roots in the complex plane.

(a) $z^5 = -32$. (b) $z^4 = -2\sqrt{3} - 2i$.

10. By using De Moivre's theorem, prove that

(a) $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$; $\sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta$;

(b) $\sin^3 \theta = \frac{3}{4} \sin \theta - \frac{1}{4} \sin 3\theta$.

11. (i) Find the Cartesian equation of the locus of the complex number z satisfying each of the following:

(a) $\left| \frac{z-3}{z+1} \right| = 2$

(b) $\left| \frac{z+i}{z+2-i} \right| = 1$

(c) $z \bar{z} = 9$

(ii) If $|z| = 2$, find the locus of the points represented by the complex numbers $3z + 4$.